

DONGSHENG CHEN, Ph.D

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PROFESSIONAL SUMMARY

- **Full-Stack Robotics Engineer** (UPenn GRASP Lab) with **5+ years of experience** across the robot development lifecycle, specializing in mechatronic systems, fluid mechanics, and reinforcement learning (RL).
- **First-principles problem solver** who excels at tackling novel challenges, systematically decomposing complex systems into manageable, mutually exclusive components to implement targeted solutions.
- **Collaborative communicator and mentor** with a proven record of leading and coordinating an interdisciplinary team to solve complex, system-level problems across mechanical, electrical, and software domains.
- **Peer-reviewed researcher** with publications in top-tier robotics venues including **IEEE RA-L** and **ASME**.
- **Passionate roboticist** eager to leverage full-stack expertise to build, deploy, and scale innovative robotic systems in a fast-paced industry setting.

SKILLS

Robotics	Reinforcement Learning , Feedback Control (PID, LQR), Path Planning, State Estimation (Sensor Fusion, Kalman Filters), Simulation (Customized Simulator, MuJoCo), Mechanics (Rigid Body Kinematics, Dynamics, Fluid Mechanics)
Programming	Python (PyTorch), C/C++ , ROS, Linux/Shell, Git, MATLAB/Simulink, OpenCV
Systems	Embedded Systems (SAM21), Mechatronics Design , Cross-Stack Debugging
CAD & Fab	SolidWorks , 3D Printing, Laser Cutting, CNC Machining

EDUCATION

University of Pennsylvania, Philadelphia, PA

- Ph.D. in Mechanical Engineering & Applied Mechanics (GRASP lab) July 2026
 - Dissertation: Design, Performance Analysis, and Motion Control of a Jet-Propelled Soft Robotic Platform
- M.S.E. in Robotics 2019
 - Relevant Coursework: Feedback Control, Convex Optimization, Learning in Robotics

Jilin University, Changchun, China

- B.E. in Vehicle Engineering 2017

RESEARCH EXPERIENCE

Soft, Jet-propelled Underwater Robot Design, Analysis, and Motion Control May 2021 – Present
Graduate Research Assistant, University of Pennsylvania, Philadelphia, PA

System Implementation

- Engineered a 26 cm x 20 cm deformable origami exoskeleton via laser-cutting, folding, and 3D printing to serve as a fluid storage cavity that deforms to generate a 15 N compression force for jet propulsion.
- Customized a tendon-driven actuation system—integrating DC and servo motors with a custom 3D-printed miniature clutch—to control origami exoskeleton deformation, executing asymmetric actuation profiles (2 cm/s loading, 0.5 cm/s unloading) to drive cyclic jet propulsion.

- Designed a custom PCB in Altium—integrating a motor driver, IMU, current sensor, RF module, and SD card—and developed Embedded C firmware in Microchip Studio to coordinate tendon-driven actuation, log onboard sensory data at 20 Hz, and enable remote teleoperation.
- Modeled a 2-axis swivel nozzle in SolidWorks (fabricated via 3D printing), solving inverse kinematics and tuning PID controllers to achieve real-time thrust vectoring with 0.1° resolution for directional control.

Experiments and Analysis

- Engineered a constrained, linear swimming testbed utilizing an overhead camera to track the robot's trajectory for hydrodynamic performance evaluation.
- Processed trajectory data via Python (OpenCV) alongside onboard power metrics to quantify system performance, demonstrating a swimming velocity of 9.7 cm/s and a Cost of Transport (COT) of 1.06.
- Developed a static force-measurement setup—integrating a load cell, Arduino data acquisition, and custom Python scripts—to log real-time force profiles for jet force evaluation.
- Applied the Reynolds Transport Theorem to experimental load cell data, isolating a 2.0 N true jet force.

Simulation

- Formulated a 3D dynamical model in Python using Newton-Euler equations—incorporating added mass, hydrodynamic drag, pulsed-jet thrust, and surrogate models for time-dependent behavior—to establish a Reinforcement Learning (RL) training environment.
- Validated 2D planar simulation dynamics against experimental trajectory data via system identification, minimizing modeling errors to: 1 cm in translation, 5 cm in turn radius, and 10 deg/s in turn rate.
- Developed a 2D rendering environment in Gymnasium for kinematic visualization to facilitate debugging.

Motion Control

- Implemented an RL-based position controller in simulation for precise single-point and trajectory tracking, achieving simultaneous position and orientation convergence while executing a 1.5 m radius circular trajectory with a 0.7 cm average error.
- Applied domain randomization to ensure robust sim-to-real transfer—injecting latency, sensor noise, and dynamic parameter variations—outperforming baseline models by reducing average tracking error by 66% (from 15 cm to 5 cm) during continuous trajectory tasks in simulation.
- Deployed the simulation-trained control policy onto a custom hardware architecture, achieving a 66.7% success rate in zero-shot sim-to-real transfer for dynamic target tracking.

LEADERSHIP

- Led and mentored a team of 10 students (5 Master's, 5 Undergraduates) over 5 years, training them on technical skills, assigning tasks, and running weekly meetings to solve problems and keep everyone aligned, ultimately delivering an autonomous, squid-inspired underwater robot.
- Served as Head Teaching Assistant for a class of 80 students (Introduction to Robotics, MEAM 5200), coordinating with two co-head TAs on assignment content and grading, while writing homework solution code, answering online questions, and running experimental lab sessions.

PUBLICATIONS (SELECTED)

- Z. Yang, **D. Chen**, D. J. Levine, C. Sung. "Origami-Inspired Robot That Swims via Jet Propulsion." IEEE Robotics and Automation Letters, Oct. 2021.
- **D. Chen**, Z. Huang, C. Sung. "Electronics Design and Verification for Robots With Actuation and Sensing Requirements." ASME IDETC/CIE, Aug. 2023.